

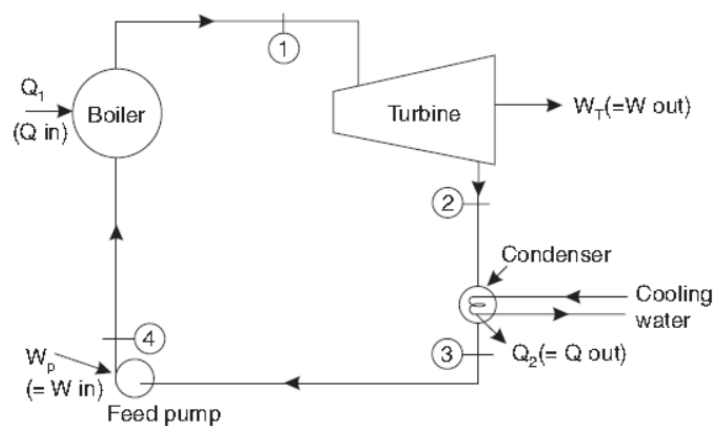
Unit-III

Power plants – Working principle of Steam, Diesel, Hydro, Nuclear power plants.
Mechanical Power Transmission-Belt Drives, Chain, Rope drives, Gear Drives and their applications.

Introduction to Robotics - Joints & links, configurations, and applications of robotics

A **power plant** is a facility that generates electric power. The most common type of power plant uses fossil fuels such as coal, natural gas, or oil to turn turbines. These turbines generate mechanical energy, which then turns into generators to produce electricity. There are also other types of power plants that use renewable energy sources such as hydroelectricity, wind power, or solar power.

Steam power plant:



Steam power plant mainly consists of four components, they are

1. Boiler
2. Steam Turbine
3. Condenser
4. Pump

Boiler may be defined as a closed vessel in which steam is produced from water by combustion of fuel. Coal is generally used as fuel in the steam power plants.

Steam Turbine: It is a device which converts kinetic energy of the steam entering in to the turbine through nozzle in to mechanical energy. Steam turbine is coupled to a generator through which power is produced.

Condenser: a condenser is a heat exchanger which condenses the low-pressure steam coming from the turbine with the external cooling medium.

Pump: it is a device which converts mechanical energy into hydraulic or pressure energy.



Working:

Water is fed into the boiler with the aid of feed water pump. Water is converted in to high pressure steam in the boiler by the combustion of coal. The high pressure and high temperature steam enter in to the steam nozzle where pressure energy of the steam is converted in to kinetic energy. High velocity steam leaving the nozzle enters in to the steam turbine where kinetic energy of the steam is transferred to the blades of the turbine and turbine is coupled to the generator to produce electricity. After expansion of steam in the turbine its pressure energy will be reduced and then low-pressure steam enters into the condenser. In the condenser steam is condensed in to water with the help of external cooling medium. Water from the condenser is drawn through the pump and sent to the boiler and then cycle repeats.

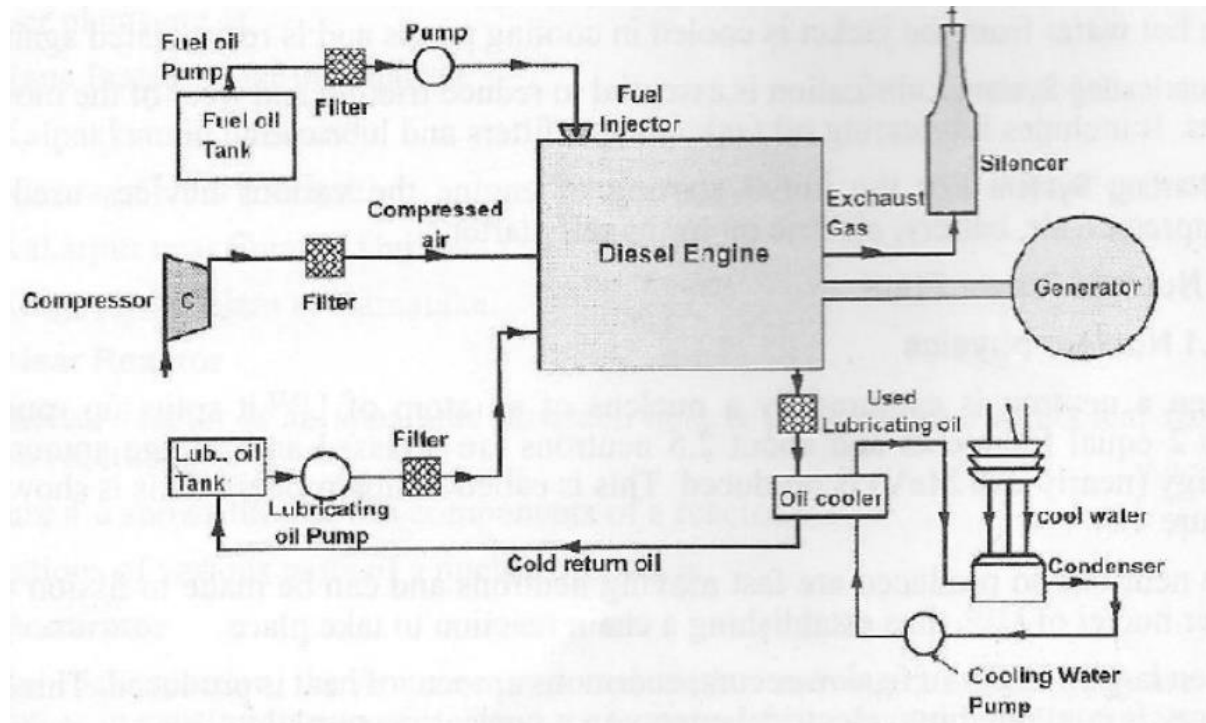
Advantages of Steam Power Plant (Thermal plant)

- Life of plant is more compared to Diesel plant
- Repair and maintenance cost is low when compared to diesel plant.
- Initial cost is less compared to nuclear plant.
- Suitable for varying load conditions.
- No radioactive harmful wastes are produced
- Unskilled operators can operate the plant.
- The power generation does not depend on the water storage.
- There are no transmission losses, as they are located near load centres.

Disadvantages of thermal power plant:

- Less efficient than diesel plants.
- Starting up and bringing into service takes more time.
- Cooling water required is more.
- Space required is more.
- Storage required for the fuel is more.
- Ash handling is a big problem
- Not economical in areas which are remote from coal fields.
- Manpower required is more.
- For large units, the capital cost is more.

DIESEL POWER PLANT



Working of diesel power plant:

Air from atmosphere is drawn into the compressor and is compressed. The compressed air is sent to diesel engine through filter. In the filter, dust, dirt from air are filtered and only clean air is sent to diesel engine. Fuel oil from tank is passed through filter where it gets filtered and clean oil is injected into the diesel engine through fuel pump and fuel injector. Mixture of compressed air and spray of fuel oil are ignited into the engine and combustion takes place. The heat energy is utilized for driving the generator, which produces power.

Main components of a diesel power plant:

Fuel Supply system: It consists of fuel tank, fuel filter and fuel pump and injector.

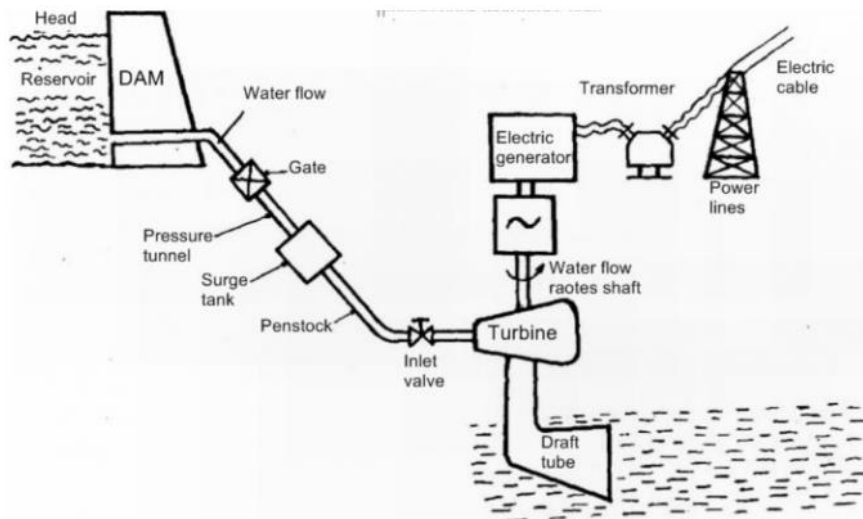
Air Intake and Exhaust system: It consists of compressor, filter and pipes for the supply of air and pipes for exhaust gases. In the exhaust system silencer is provided to reduce the noise.

Cooling system: Circulates water around the Diesel engines to keep the temp at reasonably low level.

Lubricating system: It includes lubricating oil tank, pump, filters and lubricating oil.

Starting system: For initial starting the devices used are compressed air, battery, electric motor or self-starter.

Hydroelectric power plant:



Components of hydroelectric power plant:

Reservoir: Water is collected during rainy season. It is stored in the reservoir. A dam is built across the river adequate water head.

Penstock: It is a passage through which water flows from reservoir to turbine.

Surge Tank: It is installed along the penstock (between turbine and reservoir) To control or regulate the sudden water over flow and to protect the penstock from bursting. It reduces the pressure and avoids damage to the penstock due to the water hammer effect. When the load on the turbine is decreased there will be a back flow, which causes increase or decrease in pressure. It is known as water hammer.

Power House: It is building that houses that water turbine, generator, transformer and control room.

Water Turbine: Water turbines such as Pelton, Kaplan and Francis are used to convert pressure and kinetic energy of flowing water into mechanical energy.

Draft Tube: It is connected to the outlet of the turbine.

Tailrace: It refers to the downstream level of water discharged from turbine.

Generator: It is a machine used to convert mechanical energy into electrical energy.

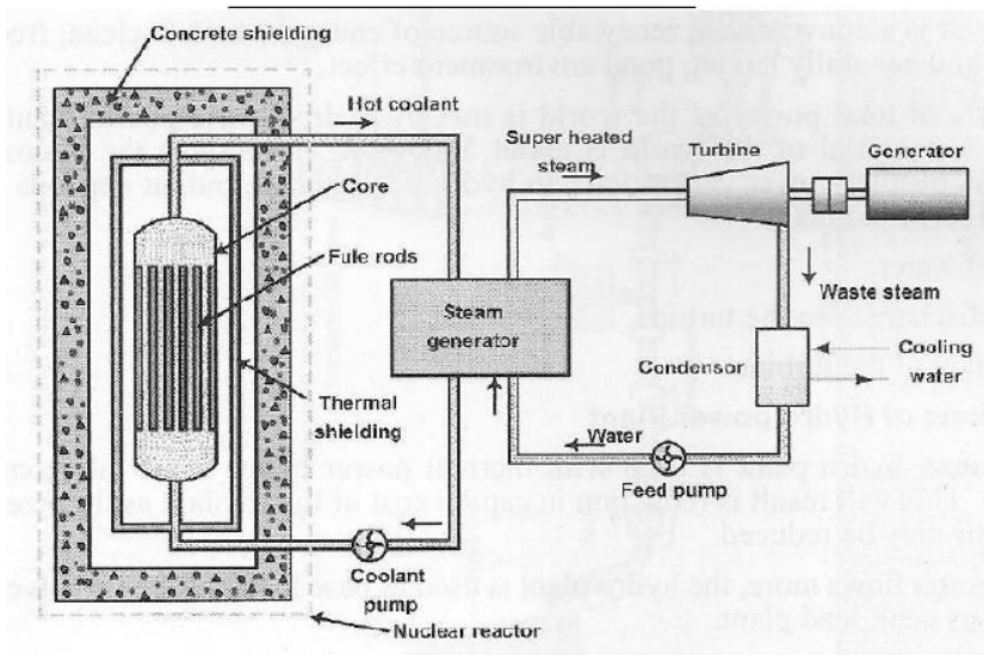
Step up transformer: It converts the Alternating Current (AC) into high voltage current suitable for transmission.

Working principle of hydroelectric power plant:

It uses the potential energy of water of water stored in a reservoir. The water from the reservoir through a penstock and then forced through nozzle or nozzles before reaching the

turbine. The hydraulic turbine converts the kinetic energy of water under pressure into mechanical energy. The shaft of the turbine is coupled to a generator that generates electricity. The electricity generated is fed to the step-up transformer to increase its voltage. Power is fed to the transmission lines for distribution. The output power of Hydel power plant depends on the head of water stored in the reservoir and the quantity of water discharged.

Nuclear power plant:



Components of nuclear power plant:

Nuclear Fuel: Normally used nuclear fuel is uranium (U_{235})

Fuel Rods: The fuel rods hold nuclear fuel in a nuclear power plant.

Neutron Source: A source of neutron is required to initiate the fission for the first time. A mixture of beryllium with plutonium is commonly used as a source of neutron.

Reactor: Nuclear fission takes place in the reactor only. Nuclear fission produces large quantity of heat. The heat generated in the reactor is carried by coolant circulated through the reactor.

Control Rods: They are used to control the chain reaction. They are absorbers of neutrons. The commonly used control rods are made up of cadmium or boron.

Moderator: Moderators are used to slow down the fast neutrons. Ordinary or heavy water are used as moderators.

Fuel Rods: The fuel rods hold nuclear fuel in a nuclear power plant.

Neutron Reflectors: To prevent the leakage of neutrons to large extent.

Shielding: To protect from harmful radiations the reactor is surrounded by a concrete wall of thickness about 2 to 2.5 m.



Working principle of Nuclear Power Plant:

- The heat generated in the reactor due to the fission of the fuel is taken up by the coolant. The hot coolant then leaves the reactor and flows through the steam generator.
- In the steam generator the hot coolant transfers its heat to the feed water which gets converted into steam.
- The steam produced is passed through the turbine, which is coupled with generator.
- Hence the power is produced during the running of turbine.
- The exhaust steam from the turbine is condensed in the condenser.
- The condensate then flows to the steam generator through the feed pump.
- The cycle is thus repeated.

Advantages of Nuclear Power Plant:

- Requires less space compared to steam power plant.
- Fuel required is negligible compared to coal requirement.
- Fuel transport cost is less.
- Reliable in operation.
- Cost of erection is less.
- Water required is very less.

Disadvantages of Nuclear Power Plant:

- Initial Cost is higher.
- Not suitable for varying load condition.
- Radioactive wastes are hazardous. Hence these are to be handled with much care.
- Maintenance cost is higher.
- Trained workers are required to operate the plant.

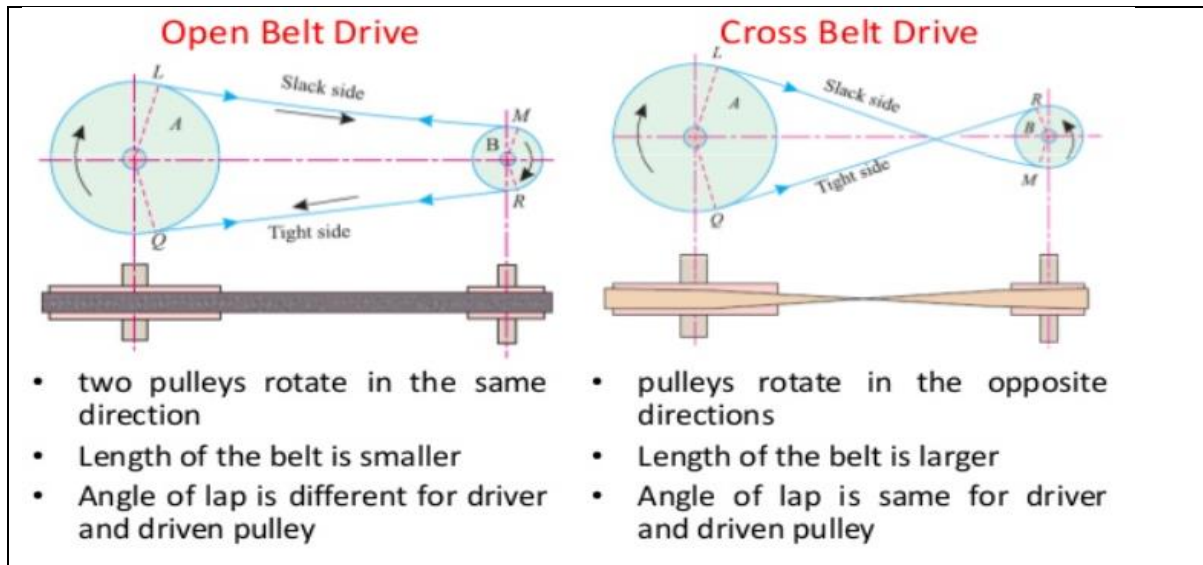
Power transmission drive: The power transmission process transmits a motion from one shaft to another with a connection between them like a belt, chain or gear. Power transmission products are commonly used in industrial automation, construction machinery, and material handling system applications.

Types of power transmission drives: There are six mechanical power transmission drives

- Belt drives.
- Rope drives.
- Chain drives.
- Gear drives.
- Shaft couplings.
- Power screws (lead screws)

Belt drive: A belt drive is a frictional drive that transmits power between two or more shafts using pulleys and an elastic belt. In most cases, it is powered by friction but it may also be a positive drive. It can operate at wide ranges of speed and power requirements. It is also highly efficient.

Types of belt drive:



Selection of a Belt Drive Following are the various important factors upon which the selection of a belt drive depends:

1. Speed of the driving and driven shafts,
2. Speed reduction ratio,
3. Power to be transmitted,
4. Centre distance between the shafts,
5. Positive drive requirements,
6. Shafts layout,
7. Space available, and
8. Service conditions.

Types of Belt Drives The belt drives are usually classified into the following three groups :

1. **Light drives.** These are used to transmit small powers at belt speeds upto about 10 m/s, as in agricultural machines and small machine tools.
2. **Medium drives.** These are used to transmit medium power at belt speeds over 10 m/s but up to 22 m/s, as in machine tools.
3. **Heavy drives.** These are used to transmit large powers at belt speeds above 22 m/s, as in compressors and generators.

Though there are many types of belts used these days, yet the following are important from the subject point of view:

1. **Flat belt.** The flat belt, as shown in Fig. 11.1 (a), is mostly used in the factories and workshops, where a moderate amount of power is to be transmitted, from one pulley to another when the two pulleys are not more than 8 metres apart.
2. **V-belt.** The V-belt, as shown in Fig. 11.1 (b), is mostly used in the factories and workshops, where a moderate amount of power is to be transmitted, from one pulley to another, when the two pulleys are very near to each other.

3. Circular belt or rope. The circular belt or rope, as shown in Fig. 11.1 (c), is mostly used in the factories and workshops, where a great amount of power is to be transmitted, from one pulley to another, when the two pulleys are more than 8 meters apart.

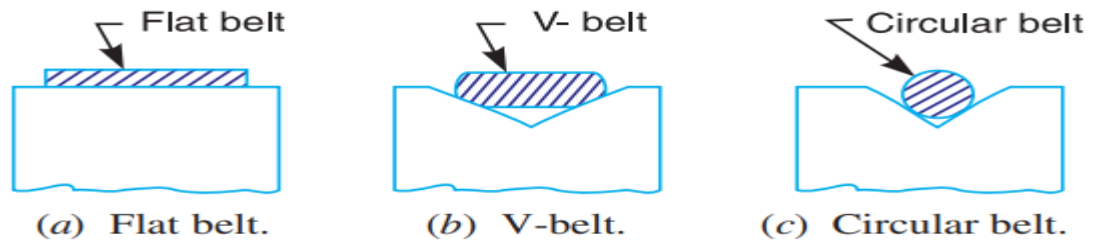
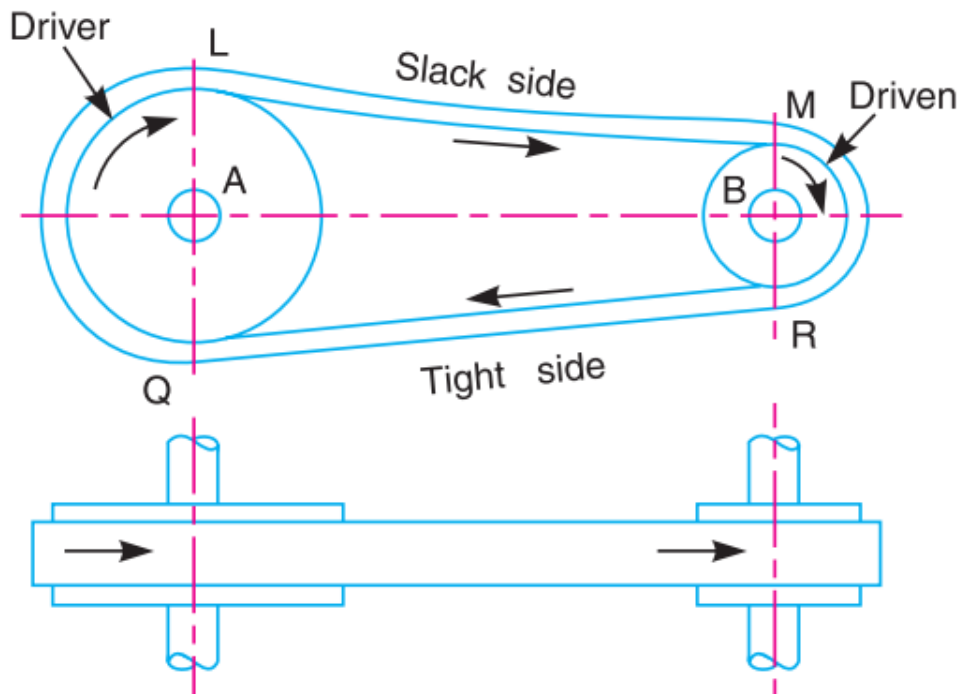


Fig. 11.1. Types of belts.

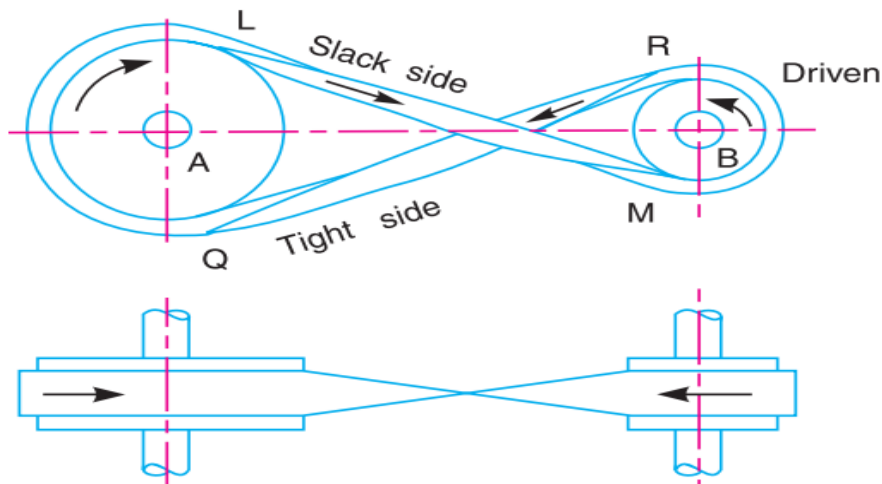
Types of Flat Belt Drives

The power from one pulley to another may be transmitted by any of the following types of belt drives: 1. Open belt drive. The open belt drive, as shown in Fig. 11.3, is used with shafts arranged parallel and rotating in the same direction. In this case, the driver A pulls the belt from one side (i.e. lower side RQ) and delivers it to the other side (i.e. upper side LM). Thus the tension in the lower side belt will be more than that in the upper side belt. The lower side belt (because of more tension) is known as tight side whereas the upper side belt (because of less tension) is known as slack side, as shown in Fig. 11.3.



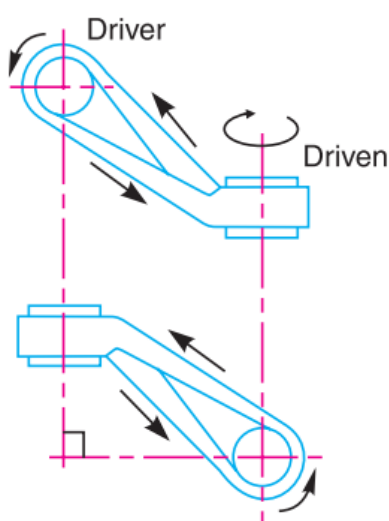
2. Crossed or twist belt drive.

The crossed or twist belt drive, as shown in Fig. 11.4, is used with shafts arranged parallel and rotating in the opposite directions.

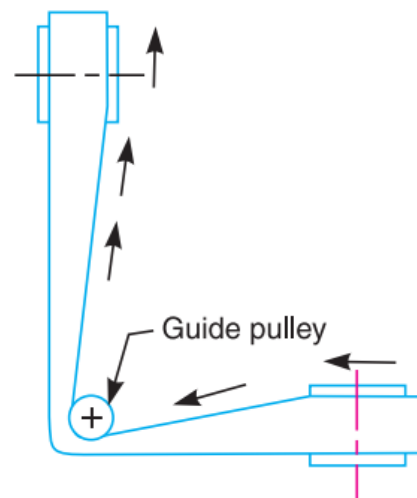


3. Quarter turn belt drive.

The quarter turn belt drive also known as right angle belt drive, as shown in Fig. 11.5 (a), is used with shafts arranged at right angles and rotating in one definite direction. In order to prevent the belt from leaving the pulley, the width of the face of the pulley should be greater or equal to $1.4b$, where b is the width of belt.



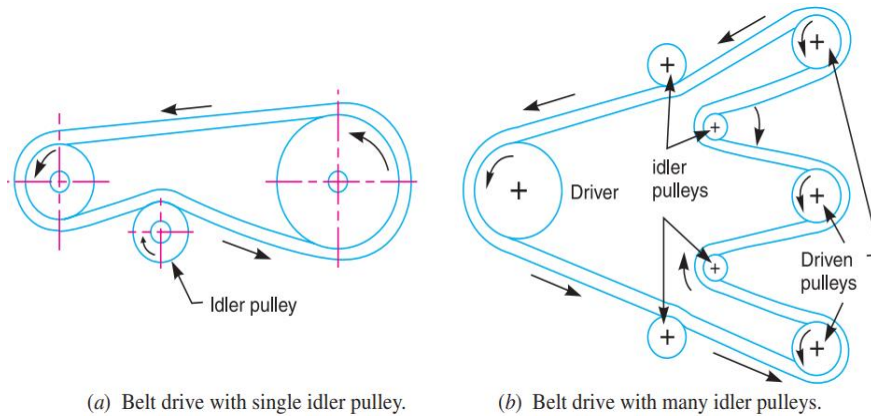
(a) Quarter turn belt drive.



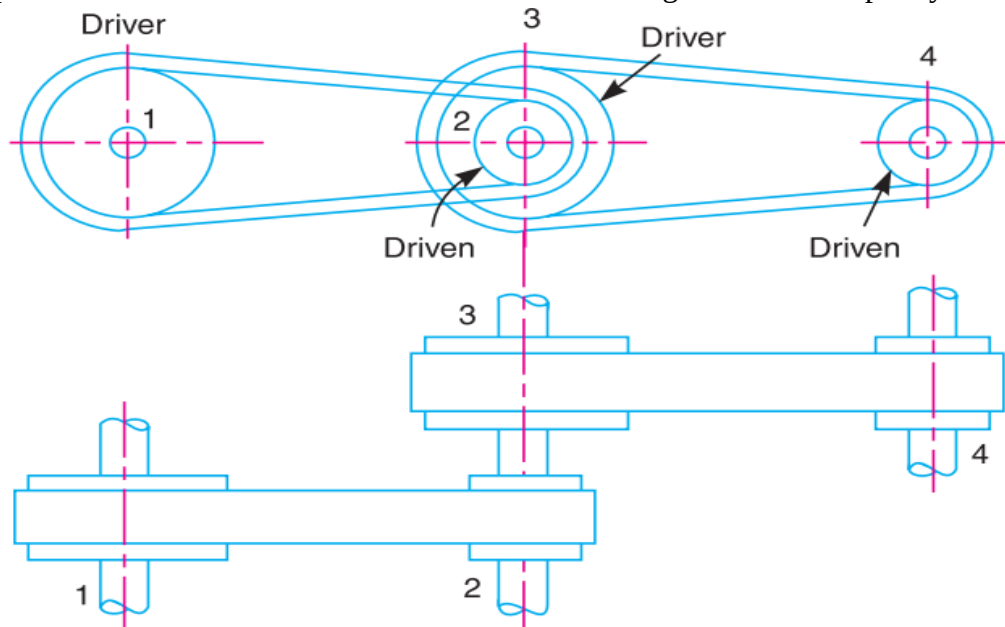
(b) Quarter turn belt drive with guide pulley.

4. Belt drive with idler pulleys.

A belt drive with an idler pulley, as shown in Fig. 11.6 (a), is used with shafts arranged parallel and when an open belt drive cannot be used due to small angle of contact on the smaller pulley. This type of drive is provided to obtain high velocity ratio and when the required belt tension cannot be obtained by other means.



1. **Compound belt drive.** A compound belt drive, as shown in Fig. 11.7, is used when power is transmitted from one shaft to another through a number of pulleys.



6. Stepped or cone pulley drive. A stepped or cone pulley drive, as shown in Fig. 11.8, is used for changing the speed of the driven shaft while the main or driving shaft runs at constant speed. This is accomplished by shifting the belt from one part of the steps to the other.

7. Fast and loose pulley drive. A fast and loose pulley drive, as shown in Fig. 11.9, is used when the driven or machine shaft is to be started or stopped whenever desired without interfering with the driving shaft. A pulley which is keyed to the machine shaft is called fast pulley and runs at the same speed as that of machine shaft. A loose pulley runs freely over the machine shaft and is incapable of transmitting any power. When the driven shaft is required to be stopped, the belt is pushed on to the loose pulley by means of sliding bar having belt forks.

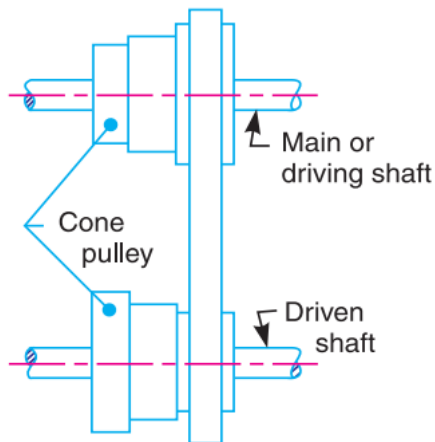


Fig. 11.8. Stepped or cone pulley drive.

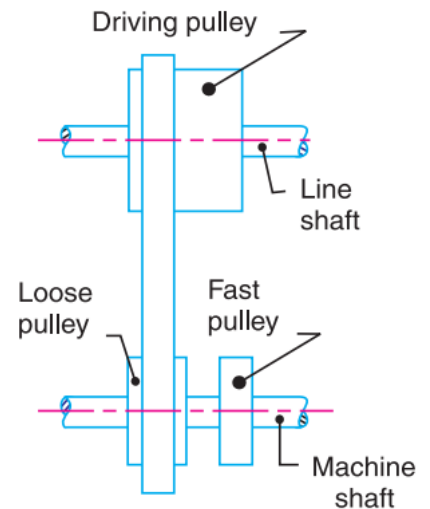


Fig. 11.9. Fast and loose pulley drive.

V-belt drive:

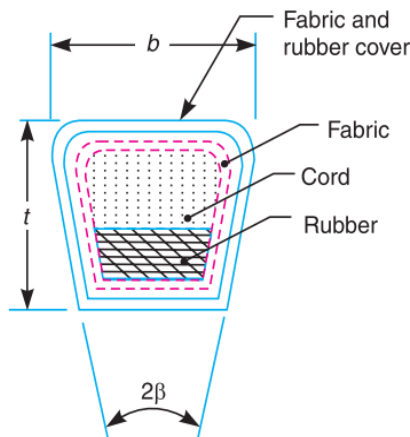
We have already discussed that a V-belt is mostly used in factories and workshops where a great amount of power is to be transmitted from one pulley to another when the two pulleys are very near to each other.

The wedging action of the V-belt in the groove of the pulley results in higher forces of friction. A little consideration will show that the wedging action and the transmitted torque will be more if the groove angle of the pulley is small.

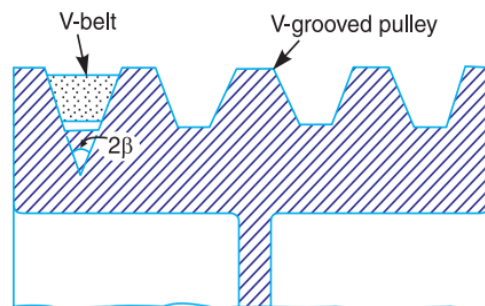
But a smaller groove angle will require more force to pull the belt out of the groove which will result in loss of power and excessive belt wear due to friction and heat. Hence a selective groove angle is a compromise between the two.

Usually, the groove angles of 32° to 38° are used. The V-belts are made of fabric and cords moulded in rubber and covered with fabric and rubber, as shown in Fig. 11.19 (a). These belts are moulded to a trapezoidal shape and are made endless. These are particularly suitable for short drives i.e. when the shafts are at a short distance apart. The included angle for the V-belt is usually from 30° – 40° .

In case of flat belt drive, the belt runs over the pulleys whereas in case of V-belt drive, the rim of the pulley is grooved in which the V-belt runs. The effect of the groove is to increase the frictional grip of the V-belt on the pulley and thus to reduce the tendency of slipping. In order to have a good grip on the pulley, the V-belt is in contact with the side faces of the groove and not at the bottom. The power is transmitted by the *wedging action between the belt and the V-groove in the pulley.



(a) Cross-section of a V-belt.



(b) Cross-section of a V-grooved pulley.

Rope Drive.

The rope drives are widely used where a large amount of power is to be transmitted, from one pulley to another, over a considerable distance. It may be noted that the use of flat belts is limited for the transmission of moderate power from one pulley to another when the two pulleys are not more than 8 metres apart.

If large amounts of power are to be transmitted by the flat belt, then it would result in excessive belt cross-section. It may be noted that frictional grip in case of rope drives is more than that in V-drive.

One of the main advantages of rope drives is that a number of separate drives may be taken from the one driving pulley. For example, in many spinning mills, the line shaft on each floor is driven by ropes passing directly from the main engine pulley on the ground floor.

The rope drives use the following two types of ropes:

1. Fibre ropes, and
2. Wire ropes.

The fibre ropes operate successfully when the pulleys are about 60 metres apart, while the wire ropes are used when the pulleys are up to 150 metres apart.

Fibre Ropes

The ropes for transmitting power are usually made from fibrous materials such as hemp, manila and cotton. Since the hemp and manila fibres are rough, therefore the ropes made from these fibres are not very flexible and possesses poor mechanical properties. The hemp ropes have less strength as compared to manila ropes. When the hemp and manila ropes are bent over the sheave (or pulley), there is some sliding of fibres, causing the rope to wear and chafe internally.

In order to minimize this defect, the **rope fibres** are lubricated with a tar, tallow or graphite. The lubrication also makes the rope moisture proof. The hemp ropes are suitable only for hand operated hoisting machinery and as tie ropes for lifting tackle, hooks etc. The cotton ropes are very soft and smooth. The lubrication of cotton ropes is not necessary. But if it is done, it reduces the external wear between the rope and the grooves of its sheaves. It may be noted that

manila ropes are more durable and stronger than cotton ropes. The cotton ropes are costlier than manila ropes.

Note: The diameter of manila and cotton ropes usually ranges from 38 mm to 50 mm. The size of the rope is usually designated by its circumference or 'girth'.

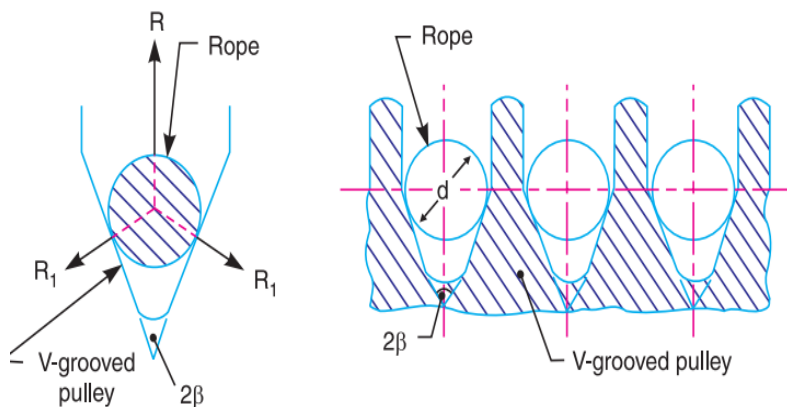
Advantages of Fibre Rope Drives

The fibre rope drives have the following advantages:

1. They give smooth, steady and quiet service.
2. They are little affected by outdoor conditions.
3. The shafts may be out of strict alignment.
4. The power may be taken off in any direction and in fractional parts of the whole amount.
5. They give high mechanical efficiency.

Sheave for Fibre Ropes

The fibre ropes are usually circular in cross-section as shown in Fig. 11.22 (a). The sheave for the fibre ropes is shown in Fig. 11.22 (b). The groove angle of the pulley for rope drives is usually 45° . The grooves in the pulleys are made narrow at the bottom and the rope is pinched between the edges of the V-groove to increase the holding power of the rope on the pulley.



(a) Cross-section of a rope.

(b) Sheave (Grooved pulley) for ropes.

Wire Ropes

When a large amount of power is to be transmitted over long distances from one pulley to another (i.e., when the pulleys are up to 150 metres apart), then wire ropes are used. The wire ropes are widely used in elevators, mine hoists, cranes, conveyors, hauling devices and suspension bridges. The wire ropes run on grooved pulleys but they rest on the bottom of the grooves and are not wedged between the sides of the grooves. The wire ropes have the following advantage over cotton ropes.

- | | |
|-------------------------------------|----------------------------------|
| 1. These are lighter in weight | 2. These offer silent operation, |
| 3. These can withstand shock loads, | 4. These are more reliable, |
| 5. They do not fail suddenly, | 6. These are more durable, |

7. The efficiency is high

8. The cost is low.

Chain Drives

We have seen in belt and rope drives that slipping may occur. In order to avoid slipping, steel chains are used.

The chains are made up of rigid links which are hinged together in order to provide the necessary flexibility for warping around the driving and driven wheels.

The wheels have projecting teeth and fit into the corresponding recesses, in the links of the chain as shown in Fig. 11.23. T

he wheels and the chain are thus constrained to move together without slipping and ensures perfect velocity ratio. The toothed wheels are known as sprocket wheels or simply sprockets. These wheels resemble to spur gears.

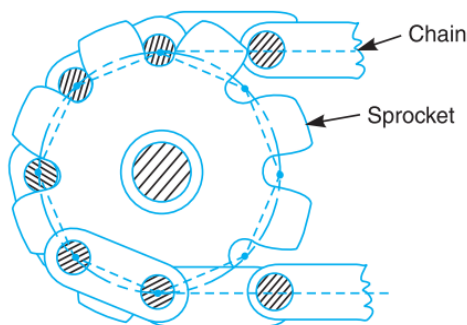


Fig. 11.23. Sprocket and chain.

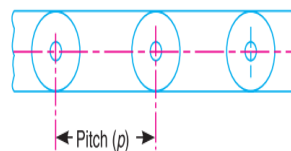


Fig. 11.24. Pitch of the chain.

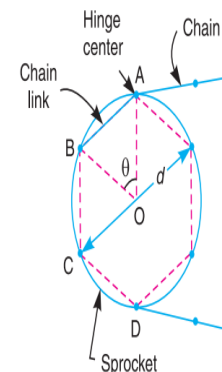


Fig. 11.25. Pitch circle diameter of the chain sprocket.

Advantages and Disadvantages of Chain Drive Over Belt or Rope Drive

Following are the advantages and disadvantages of chain drive over belt or rope drive:

Advantages:

1. As no slip takes place during chain drive, hence perfect velocity ratio is obtained.
2. Since the chains are made of metal, therefore they occupy less space in width than a belt or rope drive.
3. The chain drives may be used when the distance between the shafts is less.
4. The chain drive gives a high transmission efficiency (upto 98 per cent).
5. The chain drive gives less load on the shafts.
6. The chain drive has the ability of transmitting motion to several shafts by one chain only.

Disadvantages: 1. The production cost of chains is relatively high.

2. The chain drive needs accurate mounting and careful maintenance.

3. The chain drive has velocity fluctuations especially when unduly stretched.

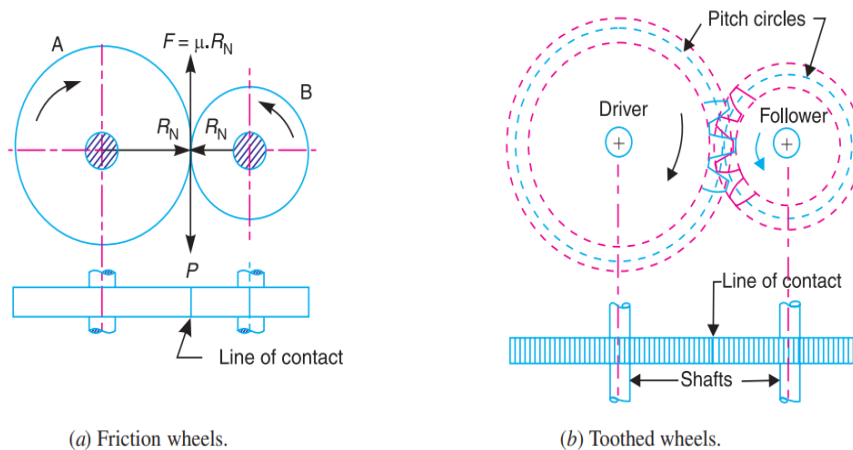
Pitch of the chain: It is the distance between the hinge centre of a link and the corresponding hinge centre of the adjacent link as shown in Fig. 11.24. It is usually denoted by p

Gear Drive:

The effect of slipping is to reduce the velocity ratio of the system. In precision machines, in which a definite velocity ratio is of importance (as in watch mechanism), the only positive drive is by means of gears or toothed wheels. A gear drive is also provided, when the distance between the driver and the follower is very small.

Friction wheels the motion and power transmitted by gears is kinematically equivalent to that transmitted by friction wheels or discs. In order to understand how the motion can be transmitted by two toothed wheels, consider two plain circular wheels A and B mounted on shafts, having sufficient rough surfaces and pressing against each other as shown in Fig. 12.1 (a).

In order to avoid the slipping, a number of projections (called teeth) as shown in Fig. 12.1 (b), are provided on the periphery of the wheel A, which will fit into the corresponding recesses on the periphery of the wheel B. A friction wheel with the teeth cut on it is known as toothed wheel or gear. The usual connection to show the toothed wheels is by their pitch circles.



Advantages and Disadvantages of Gear Drive

The following are the advantages and disadvantages of the gear drive as compared to belt, rope and chain drives:

- Advantages**
1. It transmits exact velocity ratio.
 2. It may be used to transmit large power.
 3. It has high efficiency.
 4. It has reliable service.
 5. It has compact layout.

- Disadvantages**
1. The manufacture of gears requires special tools and equipment.
 2. The error in cutting teeth may cause vibrations and noise during operation.

Robot: A robot is a machine that can perform tasks with little or no human intervention. Robots can be programmed by a computer to carry out a complex series of actions. They can be guided by an external control device, or the control may be embedded within.

Links: The rigid components that connect the joints.

Joints: The movable parts of a robot that cause relative motion between adjacent links. Joints are also known as axes.

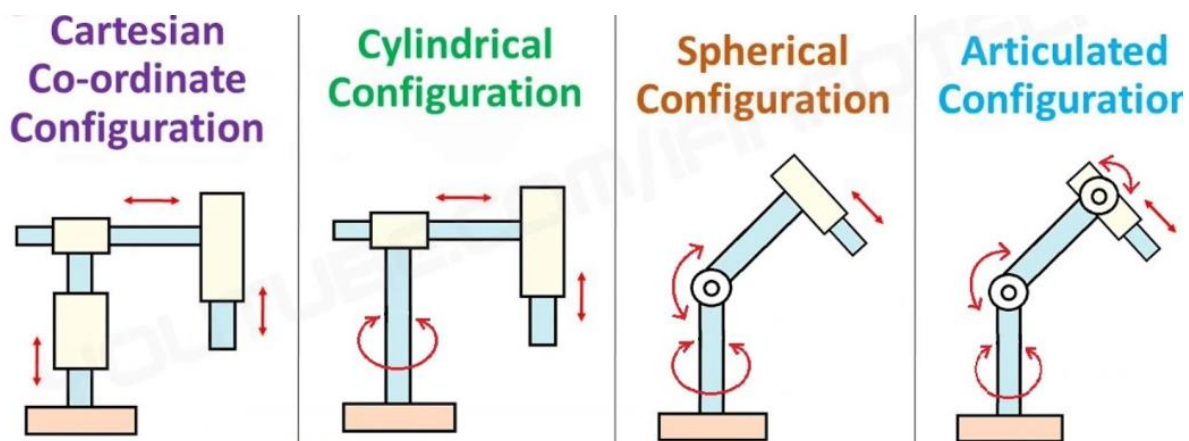
Joints are important elements in a robot that help the links travel in different kinds of movements. There are five major types of joints: Rotational joint, Linear joint, Twisting joint, Orthogonal joint, Revolving joint.

Types of joints in robot:

- Prismatic joint: A combination of a prismatic and revolute joint that allows linear motion along an axis and rotation about that axis.
- Orthogonal joint: Similar to a linear joint, but the output and input links move at right angles.
- Integrated robot joint: A cylindrical robot with a rotary joint attached to the base and one linear joint used to connect the different links.
- Spherical joint: Can move in multiple degrees of freedom around a single point.
- Revolute joint: A lower kinematic pair that enforces a cylindrical contact area.

Configuration: A robot's configuration is a specification of the position of all points of the robot. The space of all configurations of the robot is called the configuration space or C-space.

Types of configurations: There are six major types of robot configurations: Cartesian, Cylindrical, Spherical, Selective Compliance Articulated Robot Arm (SCARA), Articulate, and Delta (Parallel).

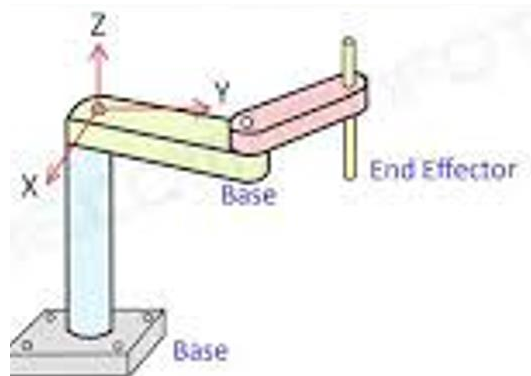


Cartesian configuration: The cartesian arm gives high precision and is easy to program. These robots are also called XYZ robots because they are equipped with three rotary joints to assemble XYZ axes.

Cylindrical configuration: Uses a vertical column and a slide that can be moved up and down along the column. The robot arm is attached to the slide so that it can be moved radially with respect to the column.

The cylinder is hollow, since there is a limit to how far the arm can retract, this creates a cylindrical dead zone around the robot structure.

SCARA configuration: The SCARA configuration is unique and designed to handle a variety of material handling operations. The SCARA's structure consists of two arms joined at the base and the intersection of arms one and two.



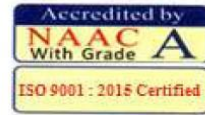
Applications of robot:

- **Material handling:** In material handling industries where, finished products are prepared for distribution, robotics is used for picking, sorting, packaging and palletizing of products.
- **Industrial robots:** Replace human workers performing simple repetitive tasks on factory assembly lines
- **Painting:** Used in automotive production and other industries to increase product quality and consistency.
- **Agriculture:** Used to plant and harvest crops, monitor crop growth, and perform other tasks to increase efficiency and reduce labor costs.
- **Construction:** Used to automate tasks such as bricklaying and concrete pouring.
- **Machine tending:** Used for overseeing a machine as it performs its task, loading and unloading machines, and other tasks.
- **Material handling:** Used to ensure goods and materials are properly stored, easy to find, or transported correctly.
- **Arc welding:** Used for automobile parts such as mufflers.
- **Robotics** is the use of computer-controlled robots to perform manual tasks. Robots are commonly used by the military and businesses to complete tasks that are dangerous for people, such as defusing bombs, exploring shipwrecks, and mines. They are also used to perform monotonous jobs, such as on an assembly line.
- The machine learning process allows robots to recognise patterns that help them understand their environment and perform specific tasks more efficiently by applying



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what they learn. By using machine learning algorithms, robots can learn autonomously without requiring specific programming for each task.

- By combining IoT with robotics, businesses can achieve industrial automation which, in turn, improves overall efficiency and cohesiveness. Both, IoT devices and robots use sensors to understand their environment, process a large volume of data, and respond accordingly.
- Intelligent robots have entered the manufacturing, healthcare, and aerospace industries. They are trained using machine learning to perform tasks like automotive assembling, powering self-driving cars, quality inspection, and medical assistance.